PRIME MINISTER'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
ARUSHA REGION

FORM THREE TERMINAL EXAMINATION

CODE:032 **CHEMISTRY**

TIME: 3: 00 HRS **01st June, 2026**

INSTRUCTIONS:

1. This paper consists of sections A, B, and C with a total of **eleven (11)** questions
2. Section **A** carries **sixteen** (16) marks, section **B** carries **fifty-four (54)** marks, and section **C** has one question which weighs **fifteen (15)** marks.
3. Answer all questions in section A and B, and **only two** questions from section C
4. Mathematical calculators may be used in the examination room.
5. **You may use the following masses where necessary:**
   H= 1, C= 12, N=14, O=16, Na= 23, Mg= 24, Al= 27, S= 32, Cl=35.5, K= 38, Ca= 40, Fe= 55.8, Cu=63.5
   - Avogadro's number = 6.022×10^{23} atoms mol^{-1}
   - GMV= 22.4 dm^3

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SECTION A: (16 MARKS)

1. Choose the letter of the correct answer and fill it in the answer sheet given.

(i) Which kind of compound is formed when two non-metals chemically combine by sharing their valence electrons?

A. Metallic compound
B. Covalent compound
C. Non-metallic compound
D. Electrovalent compound

(ii) What is the molarity of a solution containing 5.3 g of sodium carbonate in 100 mL of the solution?

A. 0.05 M
B. 0.50 M
C. 0.20 M
D. 0.01 M

(iii) A reaction in which in which reactants react to form products and then products react back to yield reactants is referred to as:

A. Synthesis
B. Endothermic
C. Irreversible
D. Reversible

(iv) Which one of the following describes a gas that burns with a ***pop sound?***

A. Supports combustion
B. Used by organisms for breathing
C. Used for filling weather balloons
D. Heaviest gas on earth

(v) A mass of calcium metal weighing 80 g was reacted with hydrochloric acid. At the end of reaction 16 g of calcium remained unreacted. How many moles reacted in the reaction?

A. 64 moles
B. 0.4 moles
C. 16 moles
D. 1.6 moles

(vi) Which of the following is the suitable method for separating mixture contents in a bottle written "**shake well before use**"?

A. Filtration
B. Sublimation
C. Evaporation
D. Distillation

(vii) The bee sting injects an acidic liquid in the blood. To relieve the acidic liquid, one should neutralize with a substance having a pH value of:

A. 2
B. 6
C. 11
D. 7

(viii) Which of the following describes a non-luminous flame?

A. Limited supply of oxygen, blue flame, noisy, three visible zones
B. Unlimited supply of oxygen, bright yellow, burns quietly, four zones
C. Limited oxygen supply, blue flame, three visible zones, noisy
D. Unlimited oxygen supply, three visible zones, noisy, blue flame

(ix) A chemical substance that is added into a base during acid-base reaction to show color change at the end of reaction is known as:

A. An indicator
B. pH scale
C. Analyte
D. Titrant

(x) It is pink in basic solution and colorless in acid solution. This statement describes which pH indicator?

A. Bromothymol blue
B. Phenolphthalein
C. Methyl orange
D. Phenol red

2. Match the items in list A with their correct responses from list B

LIST A	LIST B
(i) Quantity of the solute present in a given amount of solution.	A. Volumetric analysis
(ii) A solution which contains one mole of solute dissolved to make one litre ($1dm^3$) of the solution.	B. Titrand
(iii) A solution with precisely known concentration of a substance.	C. Molality
(iv) Concentration of a solution expressed in moles per litre of a solution.	D. Primary standard solution
(v) A chemical procedure for determining unknown concentration of solution.	E. Concentration
(vi) A solution which is always placed in the burette during acid-base titration.	F. Titrant
	G. Standard solution
	H. Molarity
	I. Molar solution

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SECTION B: (54 MARKS)

3. (a) Define the following terms as used in chemistry, give one example for each
   (i) Efflorescent salts
   (ii) Deliquescent salts
   (iii) Hygroscopic salts
   (iv) An acid
   (v) Basicity of an acid

   (b) State whether each of the following is a physical or chemical change
   (i) A paper is burnt into ashes
   (ii) A glass falls and dismantles/breaks
   (iii) Milk in the cup sours
   (iv) An iron nail turns reddish brown in color

4. (a) Give the meaning of a salt
   (b) Classify salts depending on their pH values and give the meaning of each
   (c) (i)What will happen if you immerse a red litmus paper into a beaker containing sodium hydroxide solution?
   (ii) Explain the observation that can be observed when a white copper (II) sulphate is left outside overnight.

5. (a) What does a chemical reaction mean?
   (b) By giving one balanced chemical equation for each, give the meaning of the following types of chemical reactions:
   (i) Synthesis reaction
   (ii) Precipitation reaction
   (iii) Redox reaction
   (iv) Displacement reaction

6. A chemistry student tested five solutions **A, B, C, D and E** with a universal indicator to determine their pH values. The results obtained were recorded in the table below:

Beaker of solution	**A**	**B**	**C**	**D**	**E**
pH value	**2**	**7**	**5**	**14**	**9**

   (a) Which solution was:
   (i) Strong alkali?
   (ii) Weak acid?
   (iii) Neutral solution?
   (iv) Strong acid?
   (v) Weak base?

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(b) Which beaker of solution was containing water? Give a reason
(c) Copy and complete the table below to indicate the class of fire with a correct fire extinguisher.

Burning materials	**Class of fire**	**Suitable extinguisher**
	Class C	Carbon dioxide or dry powder
Flammable liquids		
Clothing, paper and wood	Class A	
Potassium, lithium and sodium		

7. (a) The equation for formation of iron (III) chloride is as follows:

$$2Fe\ (s) + 3Cl_2 \longrightarrow 2FeCl_3$$

If the total mass of iron (III) chloride formed at STP is found to be 0.5 g. Calculate the volume of Cl_2 that reacted with iron.

(b) A hydrocarbon compound contains 79.18% carbon by mass, and rest is hydrogen. Calculate its empirical. If its vapor density is 15, calculate its molecular formula.

8. (a) Define hard water
(b) Differentiate two types of water hardness
(c) (i) Using any two balanced chemical equations, show how you can soften hardwater containing sulphate of calcium.
(ii) State three advantages of using hardwater

SECTION C: (30 MARKS)

Answer **only two** questions from this section

9. A compound **T** made of the combination between hydrogen gas and oxygen gas occupies about 71% of the earth surface. It plays a vital role in daily life including life of living things. **In six points (6),** explain the uses of the compound in daily life.

10. Acid and base react to produce water and salt. This kind of reaction has many practical applications in many fields of daily life including health, environment and soil. Explain **any six (6)** applications of this kind of reaction.

11. Chemistry experiments are carried in the laboratory. A laboratory differs from any other buildings because of its unique features. Give **six (6)** points that make a laboratory unique, and give a reason for each point.

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